THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 1: **Coordinates** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $a - b > 0$ and $ab < 0$, the point (a, b) lies in: 1

(a) Quadrant III **(b)** Quadrant II **(c)** Quadrant IV **(d)** Quadrant I

2. The perimeter of the triangle with vertices $(0, 0)$, $(12, 0)$ and $(0, 16)$ is: 1

(a) 47 **(b)** 49 **(c)** 50 **(d)** 48

3. The point $(8, -5)$ is reflected first in the x-axis and then in the y-axis. The final point is: 1

(a) $(-8, 5)$ **(b)** $(5, -8)$ **(c)** $(-8, -5)$ **(d)** $(8, 5)$

4. The area of the rectangle with opposite vertices $(0, -4)$ and $(9, 3)$ and sides parallel to the axes is: 1

(a) 63 **(b)** 64 **(c)** 62 **(d)** 65

5. The square of the distance between $(7, 1)$ and $(-2, 9)$ is: 1

(a) 150 **(b)** 145 **(c)** 142 **(d)** 147

6. The midpoint of $(4, 5)$ and $(-1, -2)$ is: 1

(a) $(2.5,\ 3.5)$ **(b)** $(1.5,\ 1.5)$ **(c)** $(2,\ 1.5)$ **(d)** $(1.5,\ 2.5)$

7. One end of a diameter of a circle is $(-6, -2)$ and the centre is $(-2, 5)$. The other end is: 1

(a) $(2, -12)$ **(b)** $(12, 2)$ **(c)** $(-2, 12)$ **(d)** $(2, 12)$

8. The point on the x-axis equidistant from $(-6, 2)$ and $(-4, -2)$ is: 1

(a) $(5, 0)$ **(b)** $(-5, 0)$ **(c)** $(0, -5)$ **(d)** $(-4, 0)$

9. If $ab > 0$ and $a + b > 0$, the point (a, b) lies in: 1

(a) Quadrant IV **(b)** Quadrant III **(c)** Quadrant I **(d)** Quadrant II

10. The perimeter of the triangle with vertices $(0, 0)$, $(15, 0)$ and $(0, 20)$ is: 1

(a) 61 **(b)** 60 **(c)** 62 **(d)** 59

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. One end of a diameter is $A(0, -8)$ and the centre of the circle is $O(5, -3)$. Find the other end B. 2

12. Find the distance between $(6, -2)$ and $(1, 10)$. 2

13. Find the midpoint of the segment joining $(1, 6)$ and $(4, -3)$. 2

14. One end of a diameter is $A(4, 10)$ and the centre of the circle is $O(-2, 3)$. Find the other end B. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that $A(-1, -2)$, $B(4, 10)$, $C(19, 46)$ are collinear. 3

16. Find the values of k for which the distance between $P(k, 0)$ and $Q(4, 8)$ is 10 units. 3

17. The vertices of a triangle are $A(-3, -3)$, $B(0, 0)$, $C(3, -3)$. Show that it is an isosceles right-angled triangle. 3

OR

Show that $A(2, 5)$, $B(16, 53)$, $C(23, 77)$ are collinear. 3

18. Find the values of k for which the distance between $P(k, 4)$ and $Q(-1, 16)$ is 15 units. 3

OR

The vertices of a triangle are $A(2, 0)$, $B(5, 3)$, $C(8, 0)$. Show that it is an isosceles right-angled triangle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The vertices of a rhombus are $A(-21, 4)$, $B(-1, -17)$, $C(19, 4)$, $D(-1, 25)$. Show that all four sides are equal and the diagonals are unequal, so that it is a rhombus and not a square. 5

OR

Show that the quadrilateral with vertices $A(-3, -2)$, $B(1, -3)$, $C(4, 2)$, $D(0, 3)$ is a parallelogram and find the midpoint of the diagonals. 5

20. The vertices of a rhombus are $A(-4, -2)$, $B(2, -10)$, $C(8, -2)$, $D(2, 6)$. Show that all four sides are equal and the diagonals are unequal, so that it is a rhombus and not a square. 5

OR

Show that the quadrilateral with vertices $A(0, 3)$, $B(4, 2)$, $C(7, 7)$, $D(3, 8)$ is a parallelogram and find the midpoint of the diagonals. 5

END OF QUESTION PAPER — Total: 40 Marks

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